

Change Point Analysis of Brent Crude Oil Prices

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What Moves Brent Crude? A Bayesian Change Point View

Brent crude has swung from single digits to nearly \$150 a barrel over the last three and a half decades. For investors, policymakers, and energy companies, the hard question is never just “did the price move” — it’s “when did the market’s behavior structurally change, by how much, and what plausibly caused it.” This report walks through a Bayesian change point analysis of daily Brent prices (20-May-1987 to Nov-2022) built for Birhan Energies to help answer that question.

Data and Approach

We worked with daily Brent spot prices, converting them to log returns ($\log(P_t) - \log(P_{t-1})$) since raw prices are non-stationary while returns are much closer to stationary — a requirement for the change point model we use. An Augmented Dickey-Fuller test on the log-return series confirms this: the test statistic was **-16.43** (p-value $\approx 2.5 \times 10^{-29}$), strongly rejecting the presence of a unit root, i.e. the return series is stationary.

We then compiled a structured dataset of 19 major events plausibly relevant to oil markets since 1987 — wars and conflicts in producing regions, OPEC/OPEC+ policy shifts, sanctions, and global economic shocks like the 2008 financial crisis and the COVID-19 demand collapse. This event dataset lets us move from “the model found a break here” to “here’s a plausible story for why.”

The Model

We used a Bayesian change point model in PyMC: a single unknown switch point τ (given a discrete-uniform prior over the time index), two regime means (μ_1 before, μ_2 after) for the log-return series, a shared volatility parameter σ , and a Normal likelihood whose mean is selected via `pm.math.switch` depending on whether each day falls before or after τ . For computational tractability we fit this to a recent window of the series (the last 2,000 trading days) rather than the full 35-year history.

What We Found

The posterior median of τ corresponds to **15 April 2020**. Comparing average prices either side of that point:

	Average Price
Before change point	\$56.63
After change point	\$73.17
Change	+29.22%

Candidate cause. The closest researched event is OPEC+’s agreement on 12 April 2020 to record coordinated production cuts (9.7 million barrels/day) — just **3 days** before the estimated change point — in response to the COVID-19 demand shock and the preceding Saudi-Russia price war. The direction is consistent with the story: after the historic supply-cut agreement, average prices in our window recovered materially relative to the crisis-era lows that preceded it.

A convergence caveat, stated plainly. MCMC diagnostics for this run show $\hat{r}$ values of 1.12 for τ and 1.08 for μ_1 — above the conventional 1.01 threshold for confident convergence, meaning the chains had not fully mixed. σ converged cleanly ($\hat{r} \approx 1.00$). In practice this means: treat the *direction and rough magnitude* of the shift as informative, but treat the *exact* date and parameter values with some caution. Re-running with more tuning samples, tighter priors, or a longer warm-up would be the first step to firm this up further.

Correlation, Not Causation

This is worth stating outright: a change point model tells us *when* the statistical properties of the price series shifted and *how large* that shift was. It does not, by itself, prove that a specific headline event *caused* the shift. Oil prices move on many simultaneous forces — currency swings, unrelated commodity moves, speculative positioning, macro data surprises. When a detected change point lines up closely with a researched event, we are proposing a **plausible, evidence-informed hypothesis** — not a proven causal claim. Full details on assumptions and limitations are documented in `docs/assumptions_and_limitations.md` in the project repository.

Why This Matters for Stakeholders

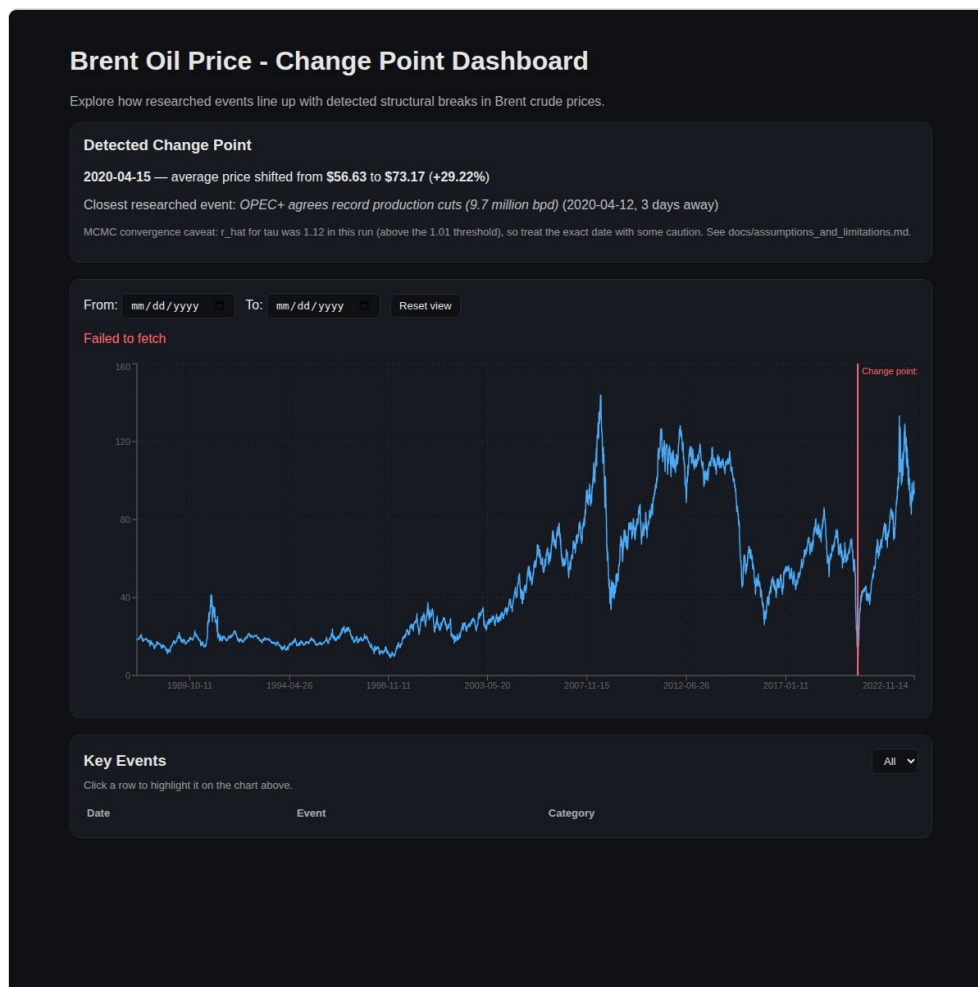
- **Investors:** structural breaks like this one mark points where prior price trends stopped being a reliable guide — useful as a trigger to re-evaluate positioning rather than extrapolate the prior regime forward.
- **Policymakers:** seeing how sharply coordinated supply decisions (like the 2020 OPEC+ cuts) can move the market underscores how much leverage production policy has over price stability.
- **Energy companies:** quantified before/after price levels support more grounded budgeting and hedging assumptions around known categories of shock (sanctions, conflict, OPEC+ policy).

Interactive Dashboard

To make these results explorable rather than static, we built a companion dashboard (Flask API + React frontend) that serves the price series, the researched events, and the change point results, with date filtering and event highlighting. See `backend_frontend_README.md` for setup instructions.

Overview — the full price series with researched events (amber lines) and the detected change point (red line) overlaid:





Event drill-down

Limitations and Future Work

- The model detects a single change point per fitted window; the full 35-year series almost certainly contains many more structural breaks (1990-91 Gulf War, 1998 price collapse, 2008 spike and crash, 2014-16 shale-driven decline, among others) that a multiple-change-point or regime-switching (Markov-switching) model would be better suited to capture.
- The current model only shifts the *mean* of returns; it does not separately model *volatility* regime shifts, which is a meaningfully different kind of market change.
- No macroeconomic controls (USD strength, global GDP growth, competing energy prices) are included; a Vector Autoregression (VAR) approach incorporating these would help disentangle overlapping drivers.
- The MCMC convergence caveat above should be resolved with a longer/re-tuned sampling run before treating specific parameter values as final.

Conclusion

Around mid-April 2020, Brent crude's price behavior shifted structurally, with average prices rising roughly 29% relative to the preceding window — a shift that lines up closely with OPEC+'s record coordinated production cuts. The broader takeaway for

Birhan Energies' clients: statistical change point detection, combined with disciplined event research, is a practical way to move from "the market moved" to "here's the most plausible story, and here's how confident we actually are in it."